

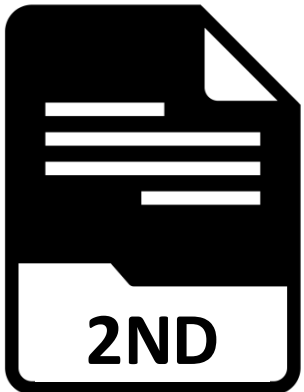
EVALUATING THE READINESS FOR EV-BASED EVACUATION

Osman Saleem, UC Santa Cruz
Dr. Clotilde Robert, UC Merced

Background

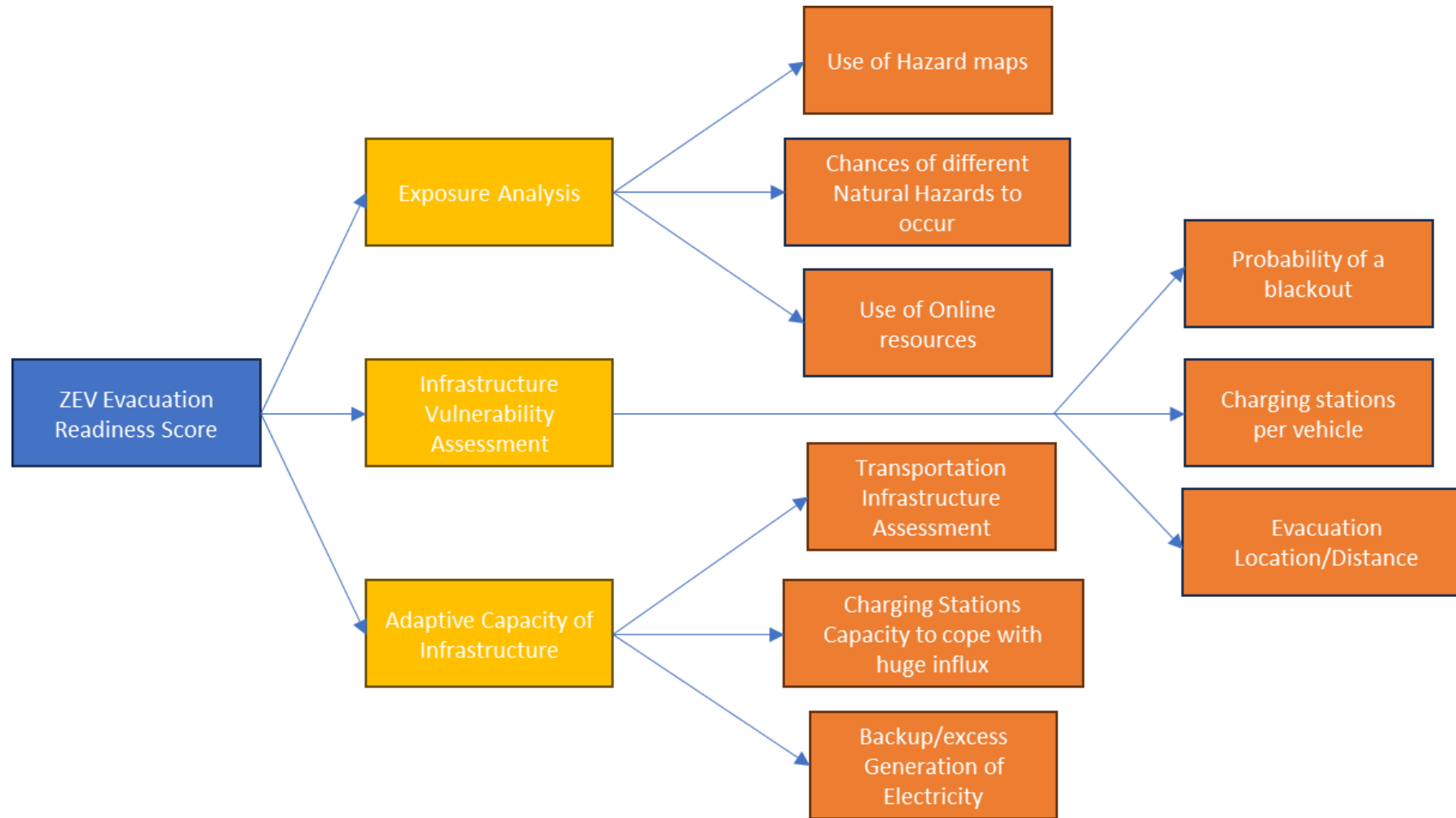


- Resilience Indicator Development
- Basic Scoring Methodology

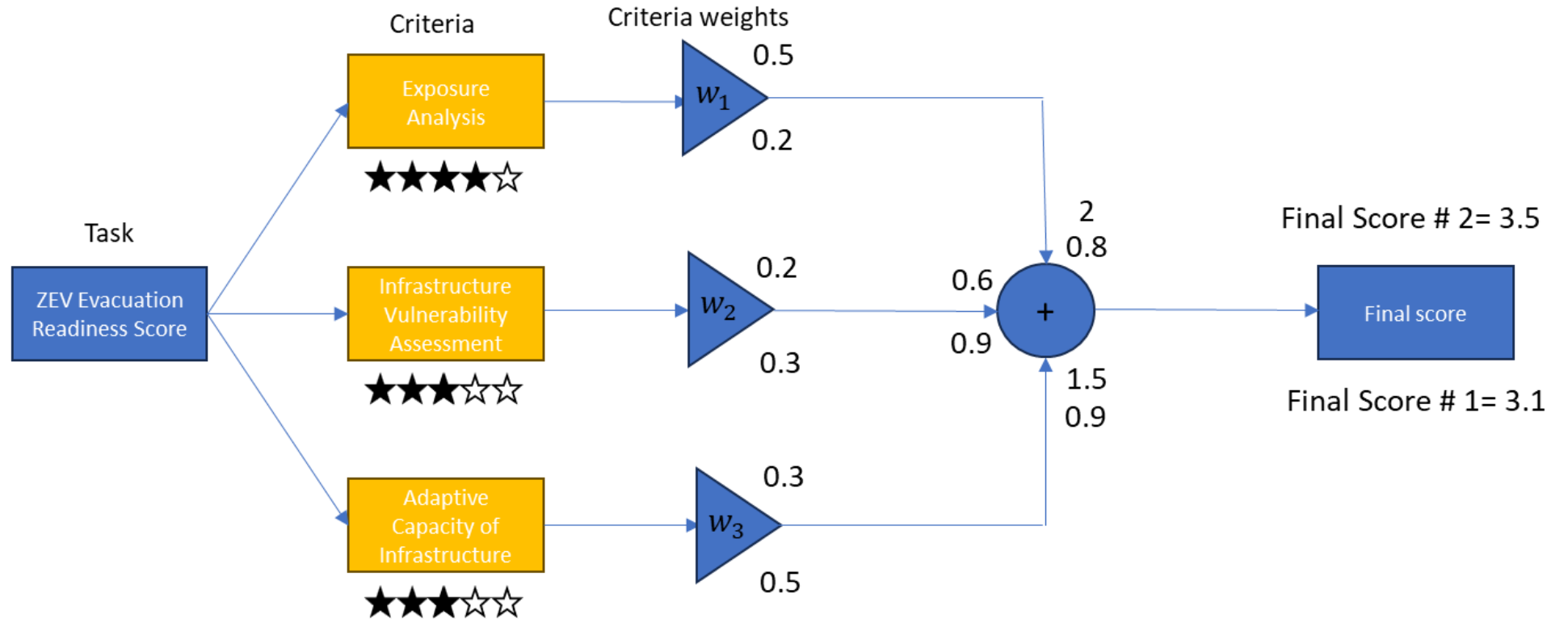


- Expanding Resilience Indicator Development (36 Indicators)
- Advanced Scoring Methodology

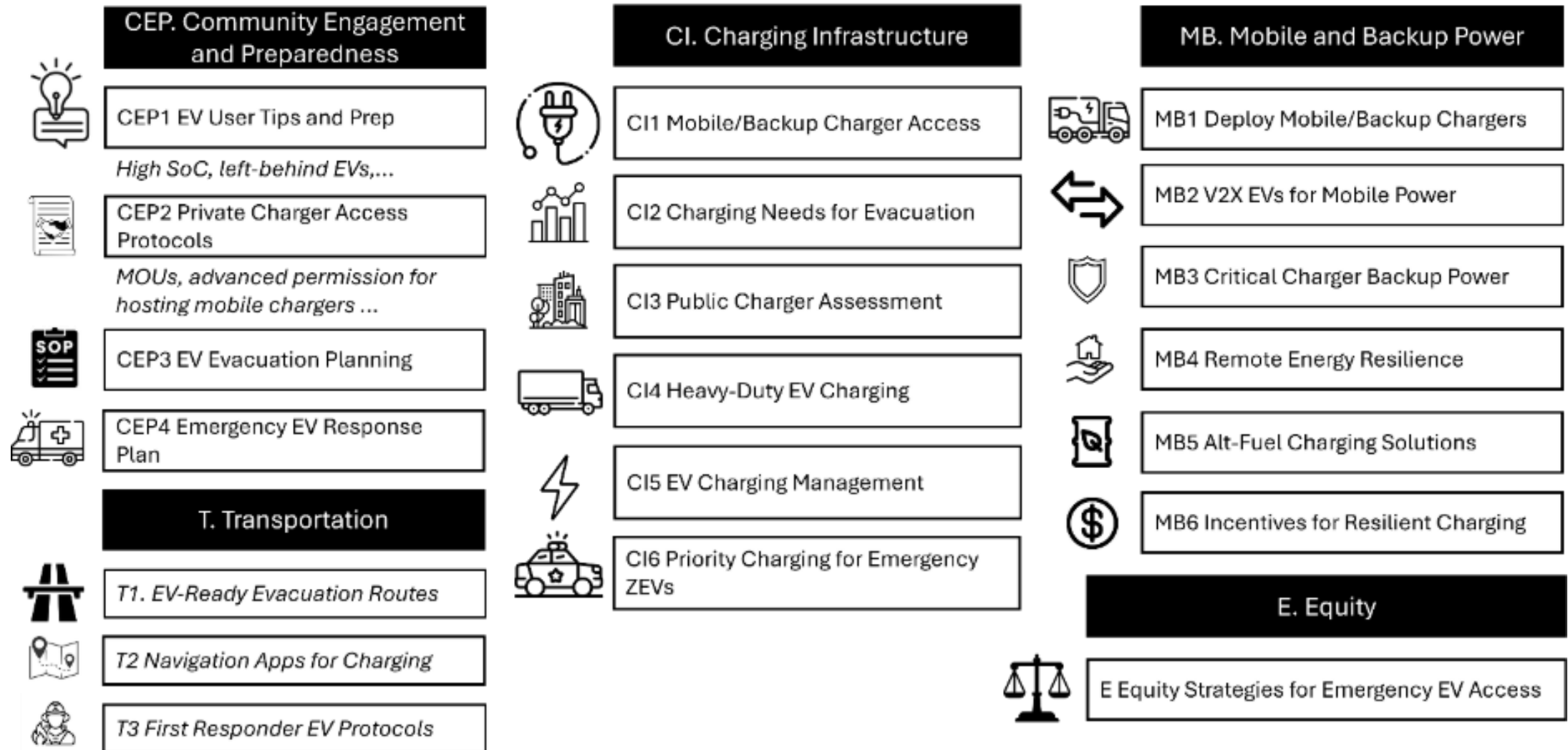
Resilience Indicators (1st Year)



Basic Scoring Methodology (1st Year)



Resilience Indicator Expansion Designed for Communities



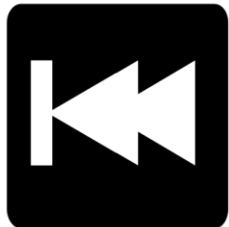
Scoring Design And Indicators Expansion

Q: Why was there a need of scoring design to get a score for the region?



Making it easier for everyone to decipher what would it mean to have the best score. Improving collaborations for making it easier for authorities to make the score better.

The aim was to develop one scoring methodology to be used by every state/region/county/city to define their own score.



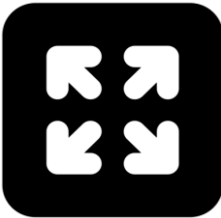
Developing a score that can be back tracked to understand which set of indicators/domains need to be revisited and improved to increase the overall score for the region.

Q: Why was there a need of reviewing the Indicators list and domains?



Some domains like "Exposure Analysis" are not applicable and only play a role in understanding the risk while the focus should be on dealing with the risk.

Developing a set of indicators spanning every domain so that a scoring methodology is not vague and captures every possible scenario in case of actual evacuation.



Making sure that these set of indicators can be implemented to any kind of temporal stage i.e. pre-disaster, during disaster and post-disaster stage.

Basic Scoring Methodology

ZEV Score Calculator

Select County:

Santa Cruz

Probability of Blackouts (hrs):

57

Number of EV Chargers:

369

Number of EVs:

10875

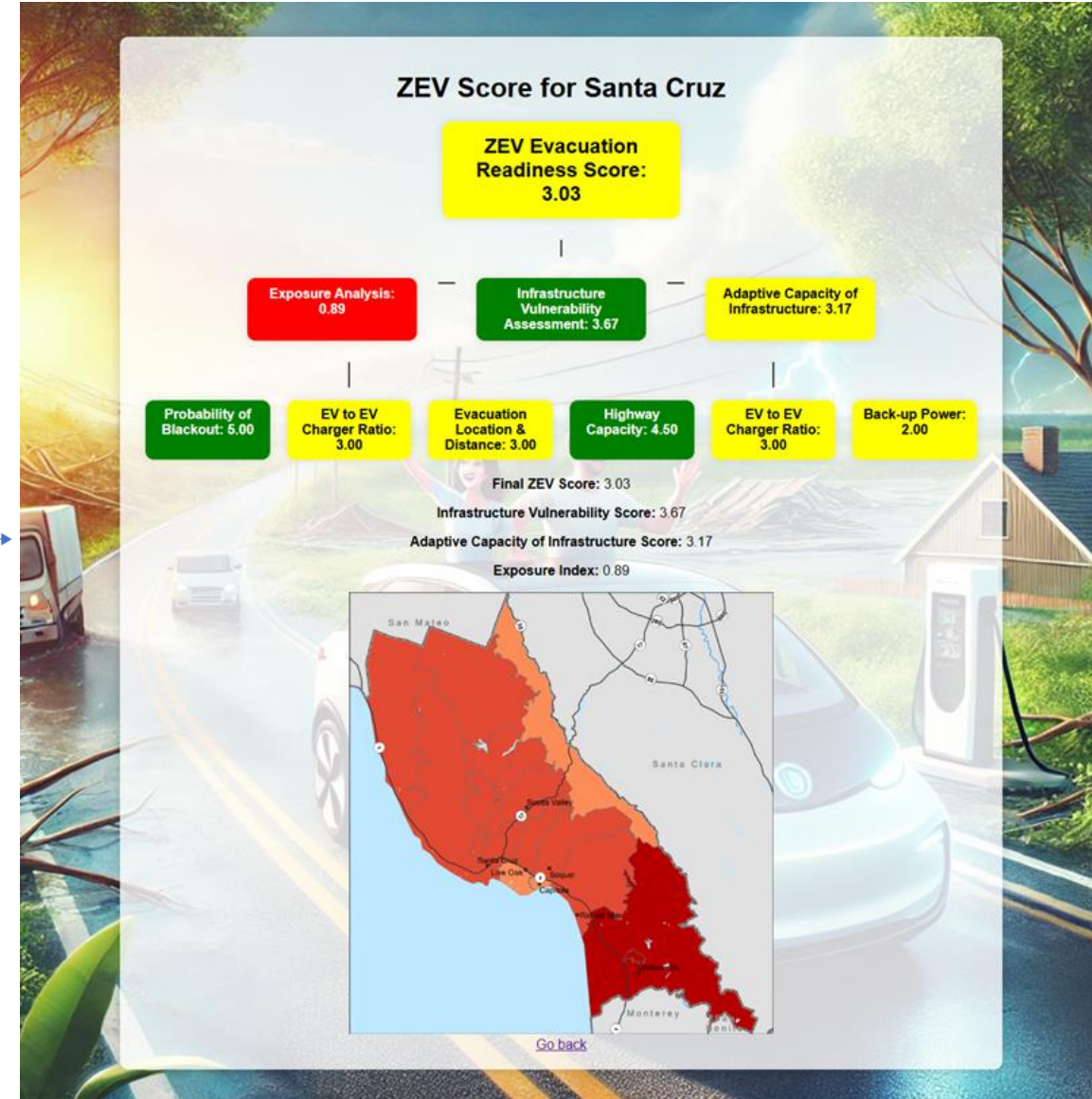
Electricity Generation (MW):

1314

Electricity Consumption (MW):

1177

Calculate ZEV Score



Scientific Resilience Indicator Expansion Design

Equity (EQ)



EQ1 : Low-income EV support



EQ2 : Vulnerable group planning



EQ3 : Mobile chargers in underserved areas



EQ4 : EVCS in disadvantaged communities



EQ5 : Multilingual EV alerts



EQ6 : Subsidized EV car-sharing/loaner

Charging Infrastructure (CI)



CI1 : EVCS emergency power backup



CI2 : EVCS surge capacity planning



CI3 : Critical site EVCS prioritization



CI4 : EVCS repair/restoration during disasters



CI5 : Temporary EVCS deployment feasibility



CI6 : Microgrid-enabled EVCS support

Community Engagement (CE)



CE1 : EV user emergency communication



CE2 : Public-private charging agreements



CE3 : Community ZEV evacuation planning



CE4 : Inter-agency emergency coordination



CE5 : Community training & awareness



CE6 : Vulnerable group outreach



CE7 : EV evacuation drills participation



CE8 : Resident feedback on EV planning

Mobile & Backup Power (MB)



MB1 : Mobile EVCS fleet availability



MB2 : EV-to-grid support planning



MB3 : Backup EVCS generator integration



MB4 : EV emergency swap stations



MB5 : EV range extension logistics

Exposure Analysis (EX)



EX1 : EV population density mapping



EX2 : EVCS location risk assessment



EX3 : EV evacuation demand hotspots



EX4 : EVCS exposure to hazards



EX5 : EV evacuation route vulnerabilities



EX6 : Power outage-prone areas with EVCS

Transportation & Routing (TR)



TR1 : EV evacuation routing system



TR2 : EV traffic congestion modeling



TR3 : EVCS routing integration



TR4 : Emergency lane prioritization for EVs



TR5 : Multi-modal EV evacuation coordination

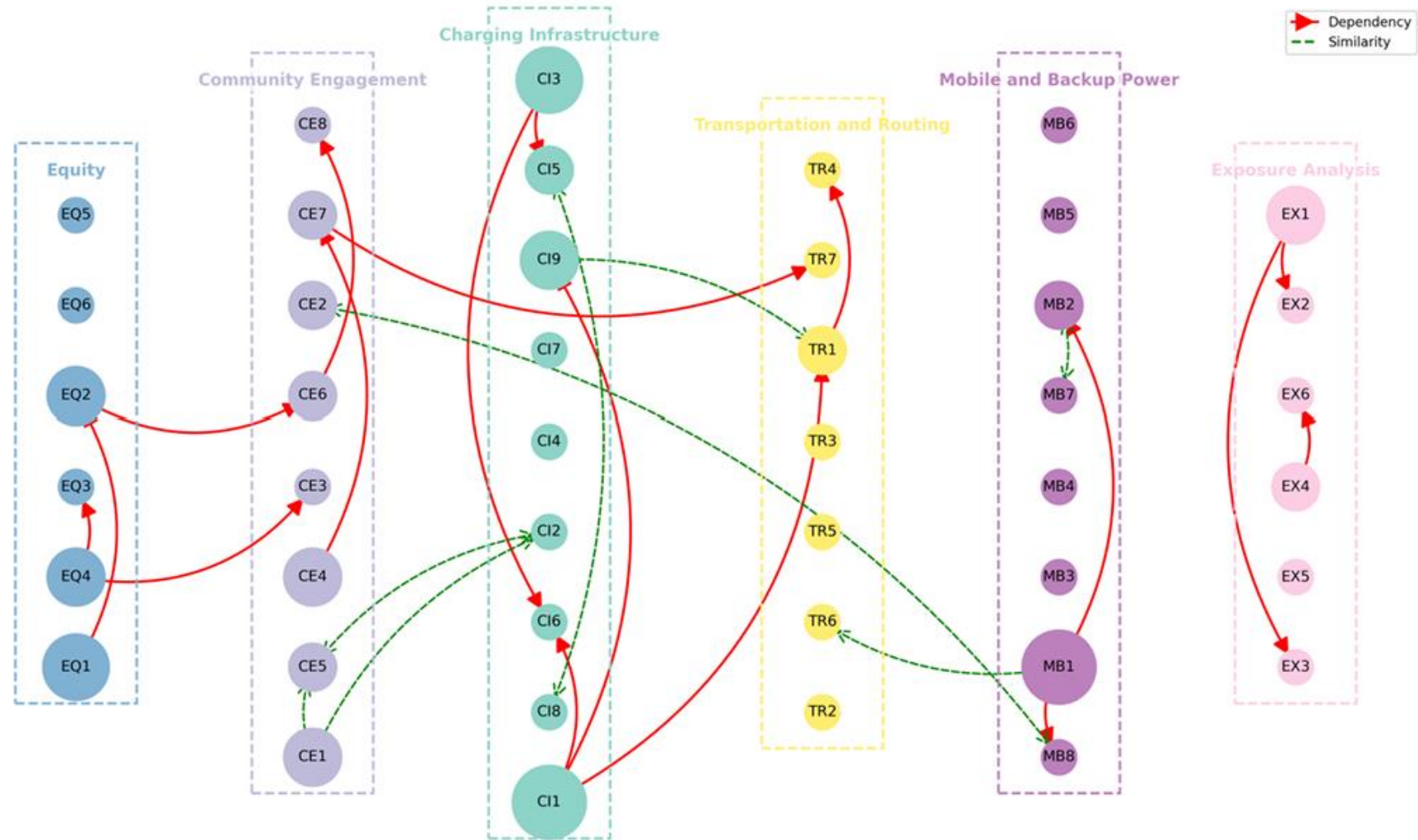


TR6 : Dynamic charging station assignment



TR7 : Traffic signal EV prioritization

Resilience Indicator cross-domain similarities & Interdependencies



**But how can we advance this research to develop an
"Advanced Scoring Design"?**

A Tool to Evaluate Readiness for EV-Based Evacuations

Dr. Clotilde Robert

Context and Goal

This work follows the report just presented by Osman.

Key question for all counties:

Are we ready to evacuate with EVs?



Objective

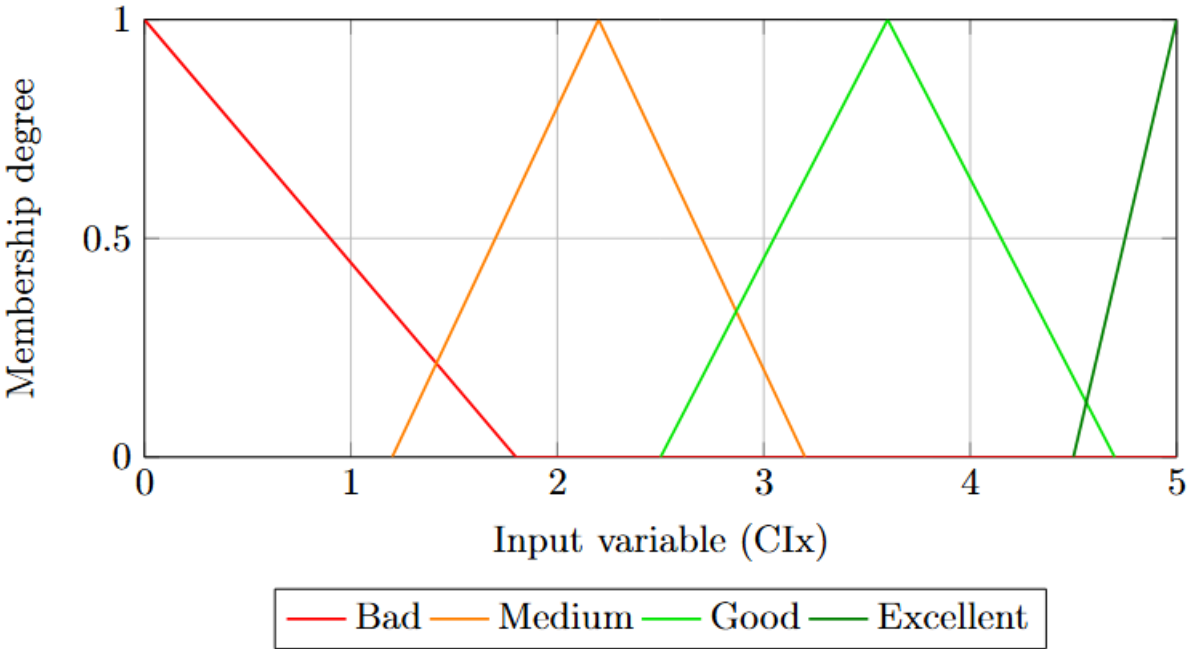
Provide communities with a way to self-assess their readiness for EV-based evacuations.

How it Works

1. Start with scoring tables

Example

Indicator C12 : Estimation of energy needs



Level	Criteria met	Description
Bad (0–1.8)	No evidence of estimation	No documented estimation of power/energy needs for EV-based evacuation. No consideration of load surges or energy constraints during evacuation.
Medium (1.2–3.2)	1 factor considered (rough estimate)	Community has begun estimating total evacuation energy needs, but the approach is limited or not scenario-based. No formal tool use or methodology. Estimates are qualitative or rely on rough calculations.
Good (2.5–4.7)	2 to 3 key factors estimated + partial use of tools or structured method	Community has quantified energy needs based on some variables (e.g., number of EVs and charging needs). Basic modeling or tool use (even if informal or partial). Results inform planning discussions but are not yet formalized.
Excellent (4.5–5)	Estimations conducted across at least 4 key factors + modeling tool used + integrated into planning	The community has estimated evacuation energy needs by scenario, including at least 4 of the following: number of EVs, evacuation distance, initial battery state, number of stations, time to evacuate. Uses tools such as EVI-X Toolbox or equivalent. Estimates are documented and considered in emergency or energy planning documents.

How it Works

2. Decision Tree

The scores are combined logically using a decision tree.

Each branch represents a step.

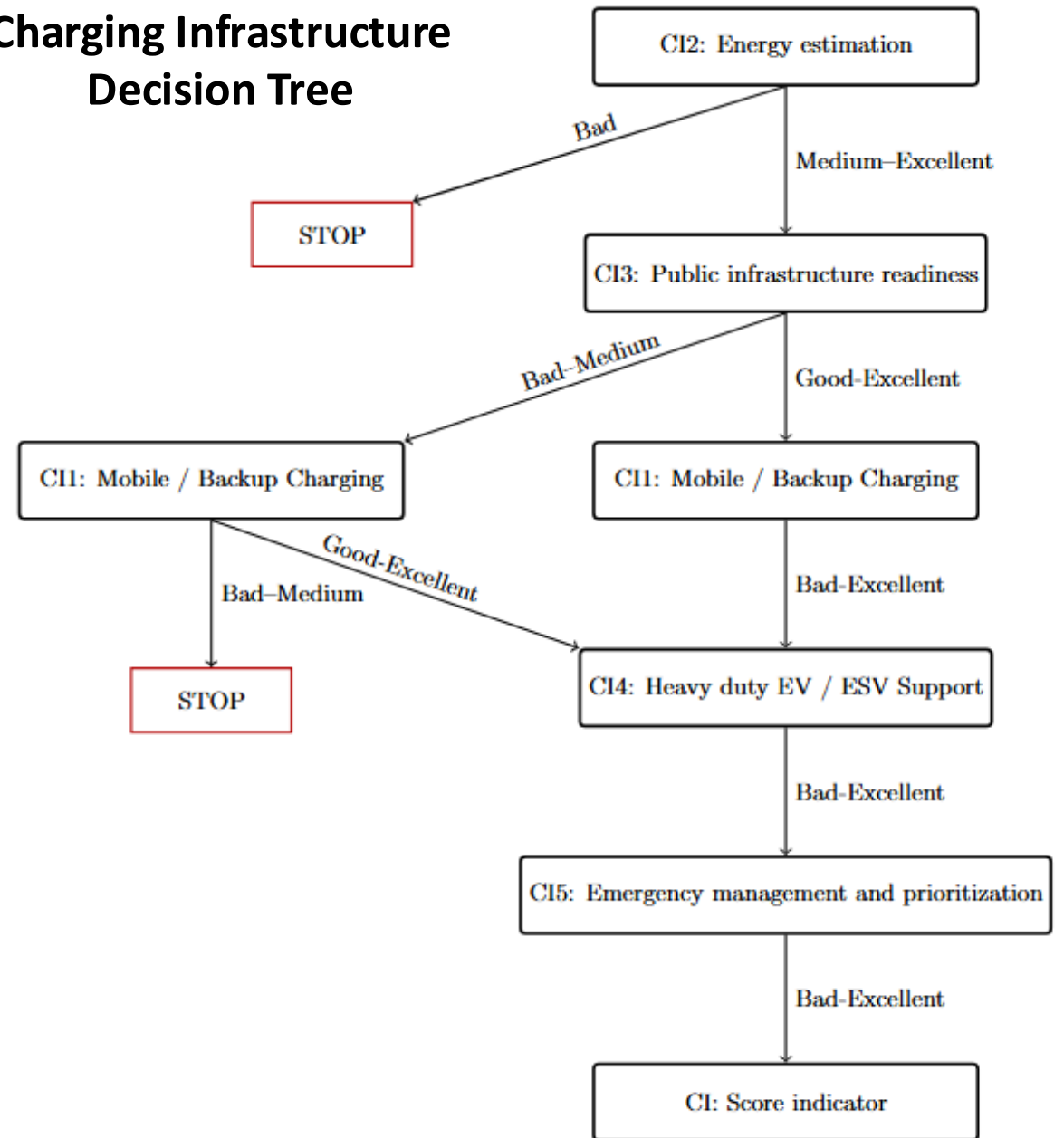
At the end: a final score between 0 and 5.

1

2

3

Charging Infrastructure Decision Tree



Case Study: Mariposa County

Example

Results overview for each indicator (0–5 scale)

CI2 – Energy needs estimation → 4/5 (Good)

Strong energy estimation and planning (see afternoon case study), but more scenarios and coordination with Mariposa County are needed to finalize planning documents.

CI3 – Public infrastructure readiness → 4/5 (Good)

Good BEV-to-charger ratio, adequate coverage, and active planning, but full implementation is still pending.

CI4 – Heavy duty / ESV support & CI5 – Emergency management and prioritization → 1.5/5 (Bad-Medium)

No heavy-duty EVs or charging protocols; planning interest but no implementation yet.

CI1 – Mobile/Backup Charging → 3.5/5 (Good)

Limited demand, candidate sites identified (see afternoon case study), but no formal agreements yet.

MATLAB R2024b - academic use

HOME PLOTS APPS EDITOR PUBLISH VIEW

File Edit View Tools Window Help

Current Folder: C:\Users\crobert\OneDrive - University of California Merced\Documents\UCM Work\ZEV Readiness Indicators\Work_Extension\MATLAB

Editor: C:\Users\crobert\OneDrive - University of California Merced\Documents\UCM Work\ZEV Readiness Indicators\Work_Extension\MATLAB\EvaluationAndQuotation.m

Command Window

Workspace

Name	Value
CI1	3.
CI2	4
CI3	4
CI4	1.
CI5	1.
fis	7x
inputs	[3
result	3.

Next Steps

This tool now needs your **expert feedback**.

*It also needs to be **expanded to other categories** beyond charging infrastructure.*



Let's move to the interactive session, we will count on your input to **evaluate and refine the scoring tables**.

Interactive session

Assessing evacuation readiness

Dr. Clotilde Robert
Osman Saleem

Instructions for the Interactive Session

Goal of the session

Collect feedback to refine and validate the indicators scoring tables

Instructions for the Interactive Session

What to do



Work in groups by **area of expertise**



Review the **scoring tables**



Discuss within your group:

- Are the current scoring ranges (e.g., 0–1.5) appropriate?
- Should they be adjusted (e.g., 0–1.2)?
- Are the level descriptions (Bad / Medium / Good / Excellent) clear and realistic?



Propose **modifications** if needed
(ranges, descriptions, recommendations)

Instructions for the Interactive Session

Example

This scoring range can be adjusted, for example it could be 0–1.

Table 8: Scoring table for CI4: Community has considered the of electric heavy-duty vehicles which are either helping in charging (charging units) or directly in evacuation

Level	Criteria met	Description
Bad (0–1.8)	No evidence of consideration or estimation	No plans or data on ESV charging needs, vehicle availability, or infrastructure designation in emergency evacuation scenarios.
Medium (1.2–3.2)	Only one factor is considered (e.g., availability of heavy-duty EVs or charger mapping)	Some preliminary steps are taken (e.g., knowing how many ESVs are in the region), but without integration into energy planning or real-time coordination.

Is this medium level difficult to reach? Maybe the current definition of medium is not appropriate.



Group 2

Propose new score from 0-1



Group 1

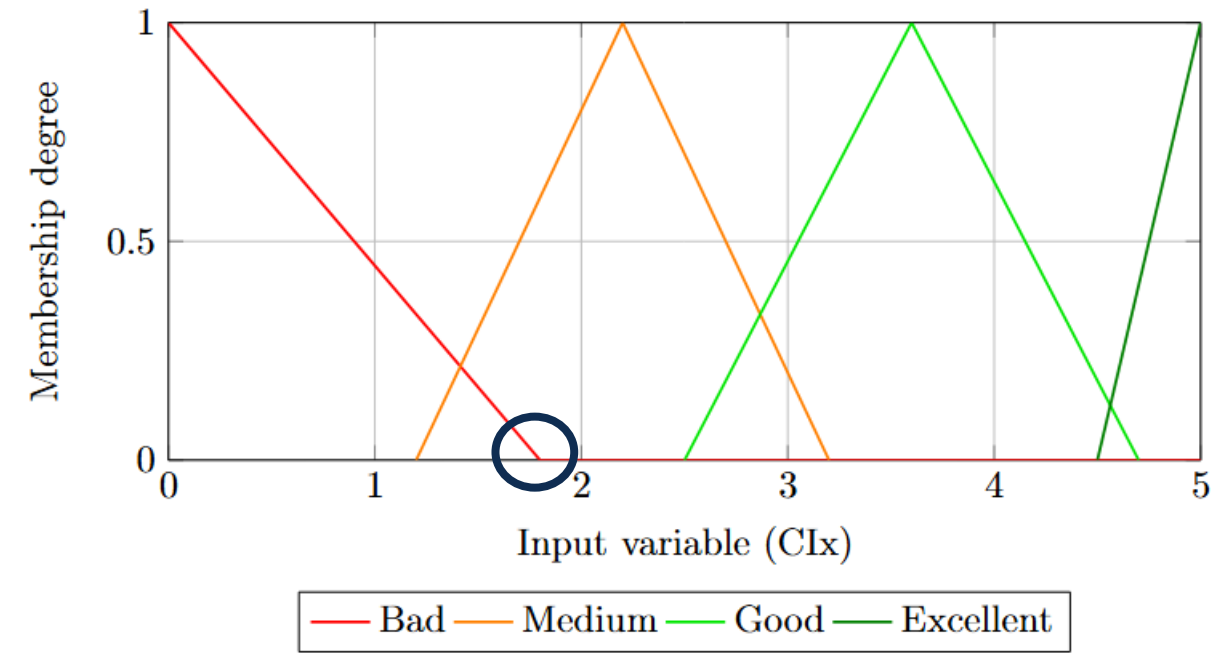
This description better fits the 'Good' level and should be moved there



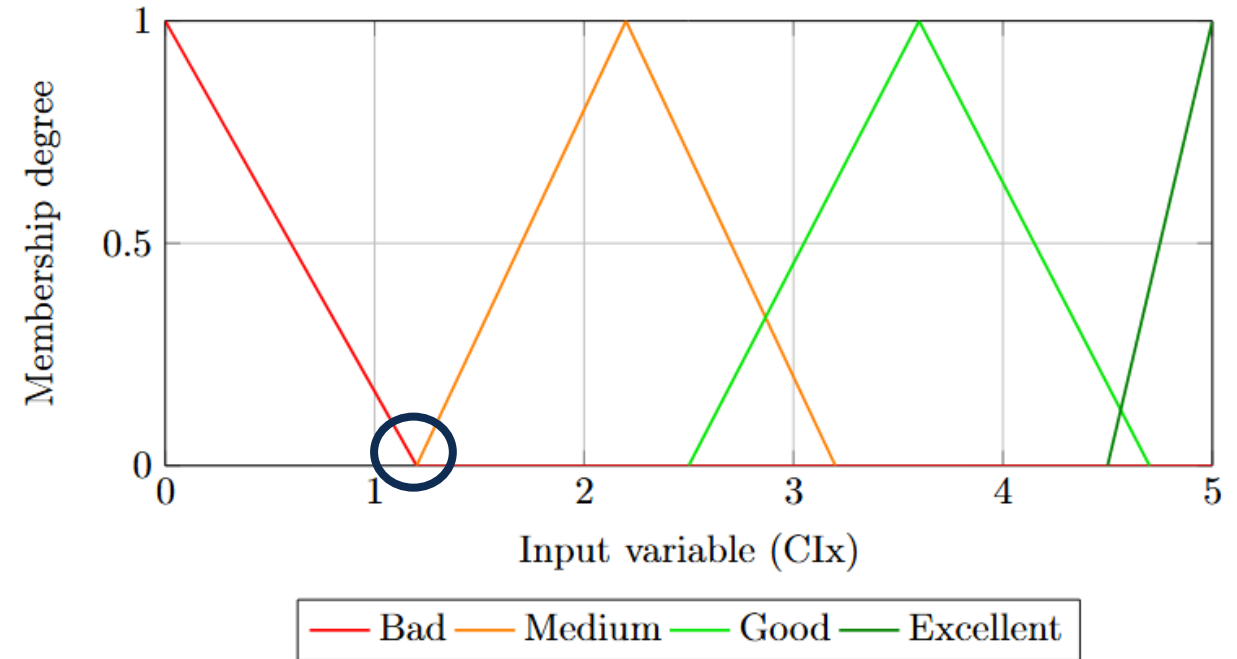
Instructions for the Interactive Session

Example

Triangular membership functions for **initial** input variable



Triangular membership functions for **new** input variable



Instructions for the Interactive Session

Materials Provided



Scoring tables for all indicators



ZEV Readiness Indicators Report



[Box Folder UC Merced](#)

Timing

1

00:20:00

Group discussion

2

00:10:00

Write down your comments (you may add notes directly on the PDF files)

3

00:05:00

Group feedback: one person per group gives a 1-minute summary

Instructions for the Interactive Session

Group Allocation

Group number	Indicator	People
1	Community Engagement & Equity	K. Corzine (UCSC); Adam (Merced County); Ricardo (UCM); Raquel (Mariposa)
2	Charging Infrastructure	Joseph (Even Recharge); Qijian (UCB); Joseph (UCM); Captain W. Smith (Mariposa)
3	Mobile and Backup Power (online session)	Bethany (Clovis); Katraman (LBL); Noah (Perimeter platform); Natalie (CARB)
4	Transportation	Wonuola (Central CAL Asthma); Saeed (UCSC); Ali (UCM); Xuchang (UCD); Sgt. Rumfelt (Mariposa)

(This allocation is a suggestion — participants may switch groups if they wish, except for the online group, which must remain fixed.)